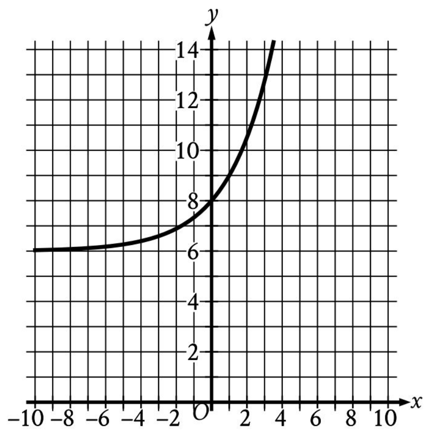
SAT 8

Math Module 1

Question #	ID
SAT8 M 1.1	f547a8b1



What is the *y*-intercept of the graph shown?

A. $(-8, 0)$

B. $(-6, 0)$

C. $(0, 6)$

D. $(0, 8)$

SAT8 M 1.2	a08ma102

The function f is defined by $f(x) = -3x + 60$. What is the value of $f(x)$ when $x = -8$?

A. 49

B. 52

C. 57

D. 84

SAT 8

Math Module 1

Question #	ID
SAT8 M 1.3	a08ma103

A producer is creating a video with a length of 70 minutes. The video will consist of segments that are 1 minute long and segments that are 3 minutes long. Which equation represents this situation, where x represents the number of 1-minute segments and y represents the number of 3-minute segments?

(A) $4xy = 70$

(B) $4(x + y) = 70$

(C) $3x + y = 70$

(D) $x + 3y = 70$

SAT8 M 1.4	a08ma104

A bowl contains 20 ounces of water. When the bowl is uncovered, the amount of water in the bowl decreases by 1 ounce every 4 days. If 9 ounces of water remain in this bowl, for how many days has it been uncovered?

(A) 3

(B) 7

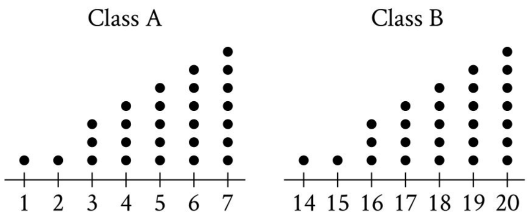
(C) 36

(D) 44

SAT 8

Math Module 1

Question #	ID
SAT8 M 1.5	d94018fd



Each of the dot plots shown represents the number of glue sticks brought in by each student for two classes, class A and class B. Which statement best compares the standard deviations of the numbers of glue sticks brought in by each student for these two classes?

A. The standard deviation of the number of glue sticks brought in by each student for class A is less than the standard deviation of the number of glue sticks brought in by each student for class B.

B. The standard deviation of the number of glue sticks brought in by each student for class A is equal to the standard deviation of the number of glue sticks brought in by each student for class B.

C. The standard deviation of the number of glue sticks brought in by each student for class A is greater than the standard deviation of the number of glue sticks brought in by each student for class B.

D. There is not enough information to compare these standard deviations.

SAT8 M 1.6 5805e747

Which expression is equivalent to $(7x^3+7x)-(6x^3-3x)$?

A. x^3+10x

B. $-13x^3+10x$

C. $-13x^3+4x$

D. x^3+4x

SAT8 M 1.7 a08ma107

In triangle ABC, the measure of angle A is 54°, the measure of angle B is 90°, and the measure of angle C is $\left(\frac{k}{2}\right)^\circ$. What is the value of k?

A. 36

B. 45

C. 72

D. 108

Question #	ID
SAT8 M 1.8	a08ma108

$$3x(x-4)(x+5)=0$$

What is one of the solutions to the given equation?

(A) -4

(B) 0

(C) 3

(D) 5

SAT8 M 1.9	a08ma109

A chemist combines water and acetic acid to make a mixture with a volume of 56 milliliters (mL). The volume of acetic acid in the mixture is $10\ \text{mL}$. What is the volume of water, in mL, in the mixture? (Assume that the volume of the mixture is the sum of the volumes of water and acetic acid before they were mixed.)

____

SAT8 M 1.10	a08ma110

A distance of 354 furlongs is equivalent to how many <u>feet</u>?
$(1 \text{ furlong} = 220 \text{ yards and } 1 \text{ yard} = 3 \text{ feet})$

(A) 306

(B) 402

(C) $25{,}960$

(D) $233{,}640$

SAT 8

Math Module 1

Question # **ID**

SAT8 M 1.11 6a87902f

$$y = 2x + 10$$
$$y = 2x - 1$$

At how many points do the graphs of the given equations intersect in the *xy*-plane?

A. Zero

B. Exactly one

C. Exactly two

D. Infinitely many

SAT8 M 1.12 a08ma112

13 is $p\%$ of 25. What is the value of p?

SAT8 M 1.13 90bcaa61

The function $f(t) = 60{,}000(2)^{\frac{t}{410}}$ gives the number of bacteria in a population t minutes after an initial observation. How much time, in minutes, does it take for the number of bacteria in the population to double?

SAT8 M 1.14 58171b5e

Each year, the value of an investment increases by 0.49% of its value the previous year. Which of the following functions best models how the value of the investment changes over time?

A. Decreasing exponential

B. Decreasing linear

C. Increasing exponential

D. Increasing linear

SAT8 M 1.15 ebbf23ae

A circle in the *xy*-plane has a diameter with endpoints $(2, 4)$ and $(2, 14)$. An equation of this circle is $(x-2)^2 + (y-9)^2 = r^2$, where r is a positive constant. What is the value of r?

SAT 8

Math Module 1

Question #	ID
SAT8 M 1.16	33e29881

In right triangle RST, the sum of the measures of angle R and angle S is 90 degrees. The value of $\sin(R)$ is $\frac{\sqrt{15}}{4}$. What is the value of $\cos(S)$?

A. $\frac{\sqrt{15}}{15}$

B. $\frac{\sqrt{15}}{4}$

C. $\frac{4\sqrt{15}}{15}$

D. $\sqrt{15}$

SAT8 M 1.17 95ed0b69

$$p = \frac{k}{4j+9}$$

The given equation relates the distinct positive numbers p, k, and j. Which equation correctly expresses $4j+9$ in terms of p and k?

A. $4j+9=\frac{k}{p}$

B. $4j+9=kp$

C. $4j+9=k-p$

D. $4j+9=\frac{p}{k}$

SAT8 M 1.18 48fb34c8

$$y > 13x - 18$$

For which of the following tables are all the values of x and their corresponding values of y solutions to the given inequality?

A.

x	y
3	21
5	47
8	86

B.

x	y
3	26
5	42
8	86

C.

x	y
3	16
5	42
8	81

D.

x	y
3	26
5	52
8	91

SAT 8

Math Module 1

Question #	ID
SAT8 M 1.19	a08ma119

Line ℓ is defined by $3y + 12x = 5$. Line n is perpendicular to line ℓ in the xy-plane. What is the slope of line n?

______

SAT8 M 1.20	a08ma120

$$y = 2(x - d)(x + d)(x + g)(x - d)$$

In the given equation, d and g are unique positive constants. When the equation is graphed in the xy-plane, how many distinct x-intercepts does the graph have?

(A) 4

(B) 3

(C) 2

(D) 1

SAT8 M 1.21	08d03fe4

For the exponential function f, the value of $f(1)$ is k, where k is a constant. Which of the following equivalent forms of the function f shows the value of k as the coefficient or the base?

A. $f(x) = 50(2)^{x+1}$

B. $f(x) = 80(2)^{x}$

C. $f(x) = 128(2)^{x-1}$

D. $f(x) = 205(2)^{x-2}$

SAT 8

Math Module 1

Question #	ID
SAT8 M 1.22	841ef26c

$$f(x) = 4x^2 + 64x + 262$$

The function g is defined by $g(x) = f(x+5)$. For what value of x does $g(x)$ reach its minimum?

A. -13

B. -8

C. -5

D. -3